

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Semestertoets 2

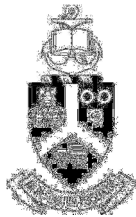
Kopiereg voorbehou

Semester Test 2

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Module EBN111
Elektrisiteit en Elektronika
17 Oktober 2015

Module EBN111
Electricity and Electronics
17 October 2015



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting: *Exam information:*

Maksimum punte: <i>Maximum marks:</i>	45	Volpunte: <i>Full marks:</i>	45
Duur van vraestel: <i>Duration of paper:</i>	90 minute <i>90 minutes</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		14	

BELANGRIK- IMPORTANT

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
2. Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All questions must be answered on the ANSWER SHEET.
3. Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
Students may do their own rough calculations on the question paper – the question paper will not be taken in.
4. Antwoorde moet eenhede insluit!
Answers must include units!
5. **Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	1 J Joubert
	2 A Chatterjee

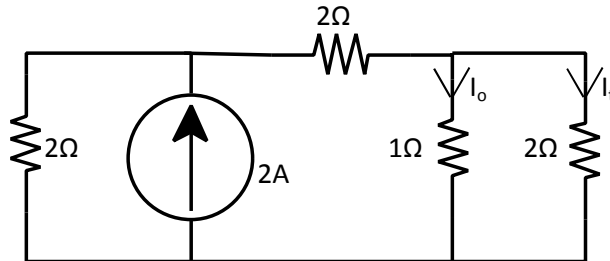
VRAAG/QUESTION 1 [23]

Vraag 1.1 [3]

- a) Aanvaar aanvanklik dat $I_t = 1\text{ A}$ en gebruik lineariteit om die regte waarde van I_o te bereken. (Nota: dit is nie 2 A nie)
- b) Bepaal I_o as die stroombron se waarde verander word na 14 A .

Question 1.1 [3]

- a) Assume initially that $I_t = 1\text{ A}$ and use linearity to find the actual value of I_o in the given circuit. (Note: it is not 2 A)
- b) Find I_o if the value of the current source is changed to 14 A .

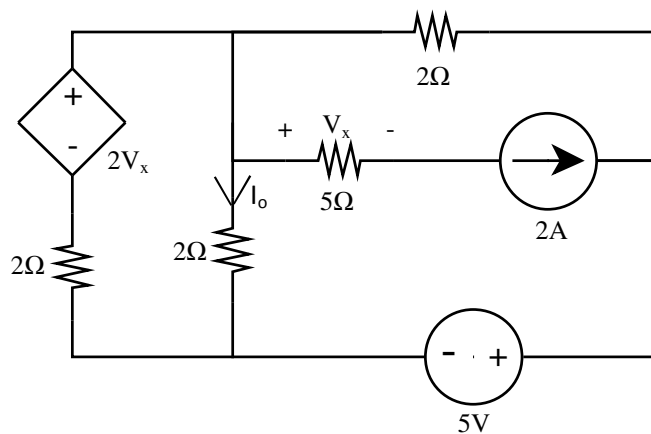


Vraag 1.2 [4]

- a) Bepaal I_o slegs agv die 5V bron.
b) Bepaal I_o slegs agv die 2A bron.

Question 1.2 [4]

- a) Find I_o due to the 5V source only.
b) Find I_o due to the 2A source only.

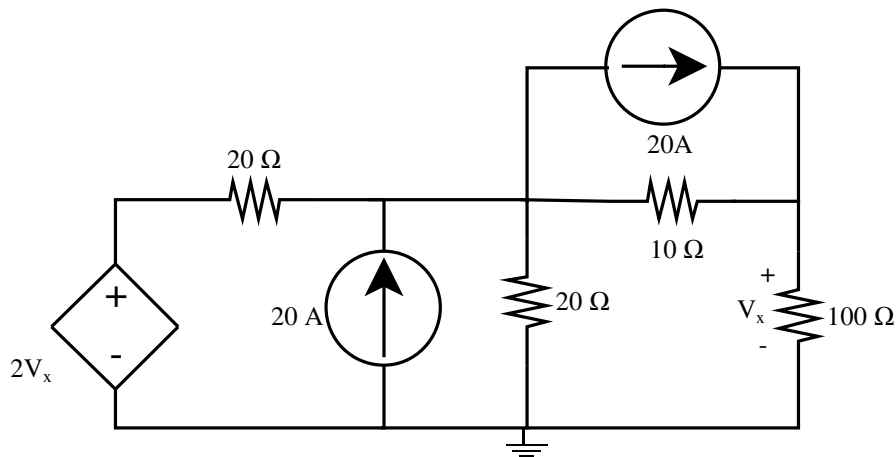
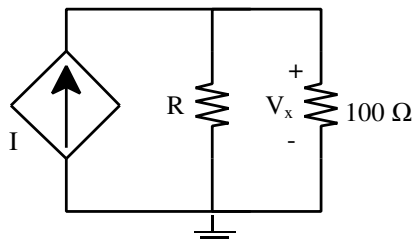


Vraag 1.3 [4]

Gebruik brontransformatie om die stroombaan in fig. 1 te transformeer na die stroombaan in fig. 2. Bepaal R en I (Wenk: I moet 'n funksie van V_x wees).

Question 1.3 [4]

Use source transformation to transform the circuit in fig. 1, to the circuit shown in fig. 2. Find R and I (Hint: I should be a function of V_x).

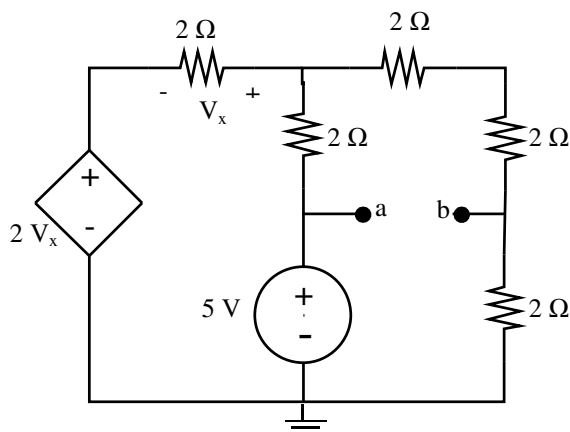
**Figure 1****Figure 2**

Vraag 1.4 [4]

- a) Bepaal die Thevenin-weerstand tov terminale a-b.
b) Bepaal die Thevenin-spanning tov terminale a-b.

Question 1.4 [4]

- a) Find the Thevenin resistance wrt terminals a-b.
b) Find the Thevenin voltage wrt terminals a-b.



Vraag 1.5 [2]

As jy 'n 20Ω las aan 'n stroombaan gekoppel word vloei daar 1A deur die las; bereken die Thevenin/Norton weerstand van die stroombaan.

Question 1.5 [2]

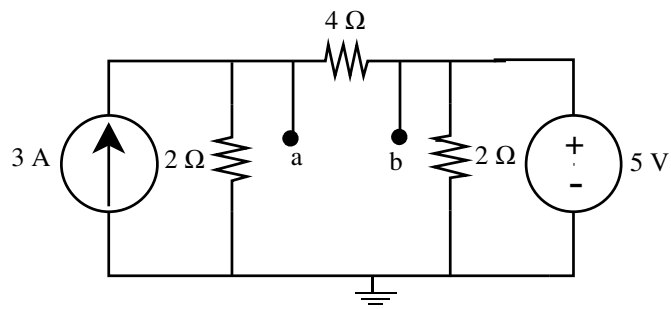
If a 20Ω load is connected to a circuit, 1A flows through this load; calculate the Thevenin/Norton resistance of the circuit.

Vraag 1.6 [4]

- a) Bepaal die Norton-weerstand tov terminale a-b.
- b) Bepaal die Norton-stroom tov terminale a-b.

Question 1.6 [4]

- a) Find the Norton resistance wrt terminals a-b
- b) Find the Norton current wrt terminals a-b.

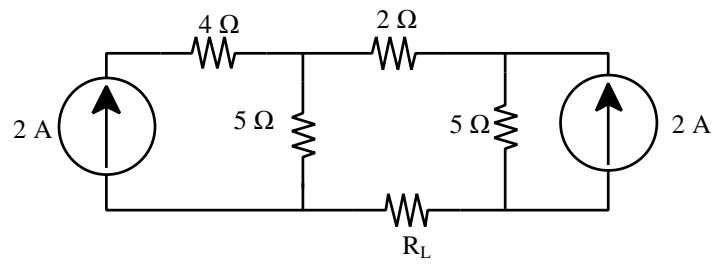


Vraag 1.7 [2]

Bepaal die waarde van die lasweerstand, R_L , wat sal verseker dat maksimum drywing oorgedra word na R_L .

Question 1.7 [2]

Find the value of the load resistor R_L that will ensure maximum power transfer to R_L .



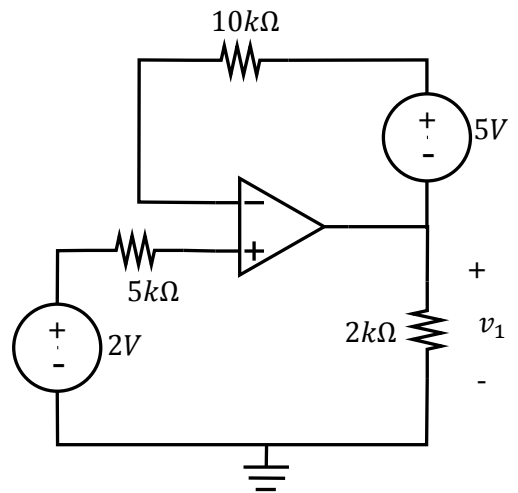
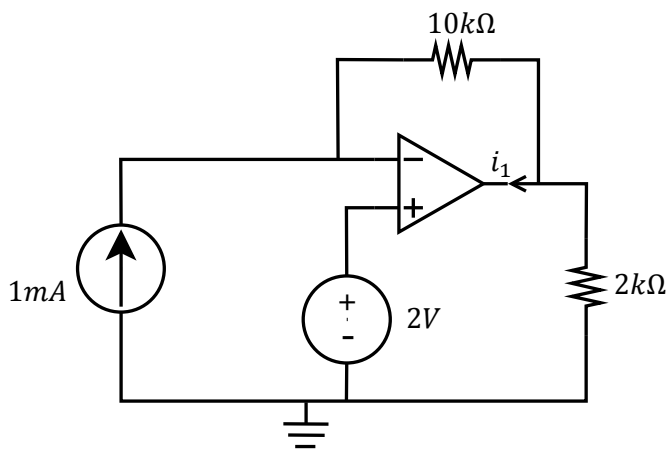
VRAAG/QUESTION 2 [22]

Vraag 2.1 [4]

Bereken i_1 en v_1 .

Question 2.1 [4]

Determine i_1 and v_1 .

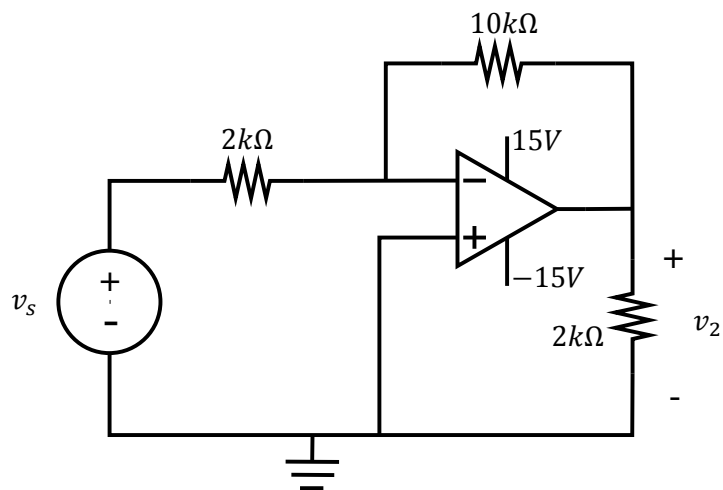


Vraag 2.2 [4]

- a) Bereken v_2 as $v_s = 2\text{ V}$.
b) Bepaal die maksimum waarde wat v_s kan wees voordat die OV in versadiging sal ingaan.

Question 2.2 [4]

- a) Determine v_2 if $v_s = 2\text{ V}$.
b) Find the maximum value of v_s before the op-amp goes into saturation.

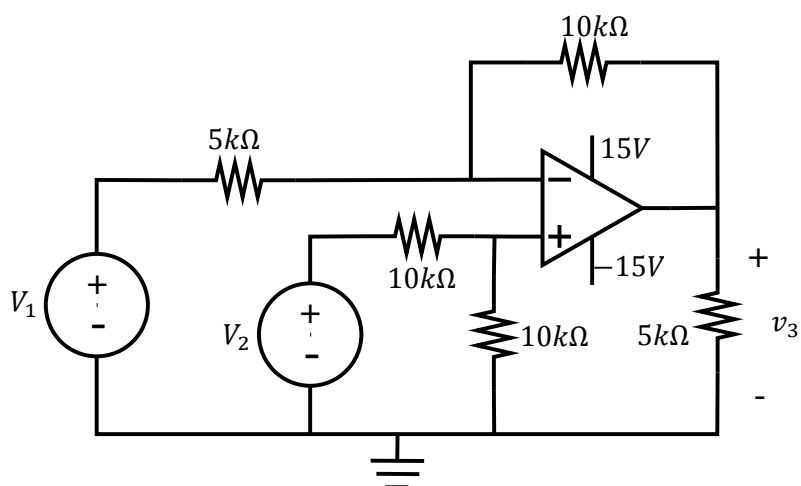


Vraag 2.3 [4]

- a) Bereken v_3 as $V_1 = 2V$ en $V_2 = -2V$.
b) Bereken v_3 as $V_1 = 0V$ en $V_2 = 10V$.

Question 2.3 [4]

- a) Determine v_3 if $V_1 = 2V$ and $V_2 = -2V$.
b) Determine v_3 if $V_1 = 0V$ and $V_2 = 10V$.

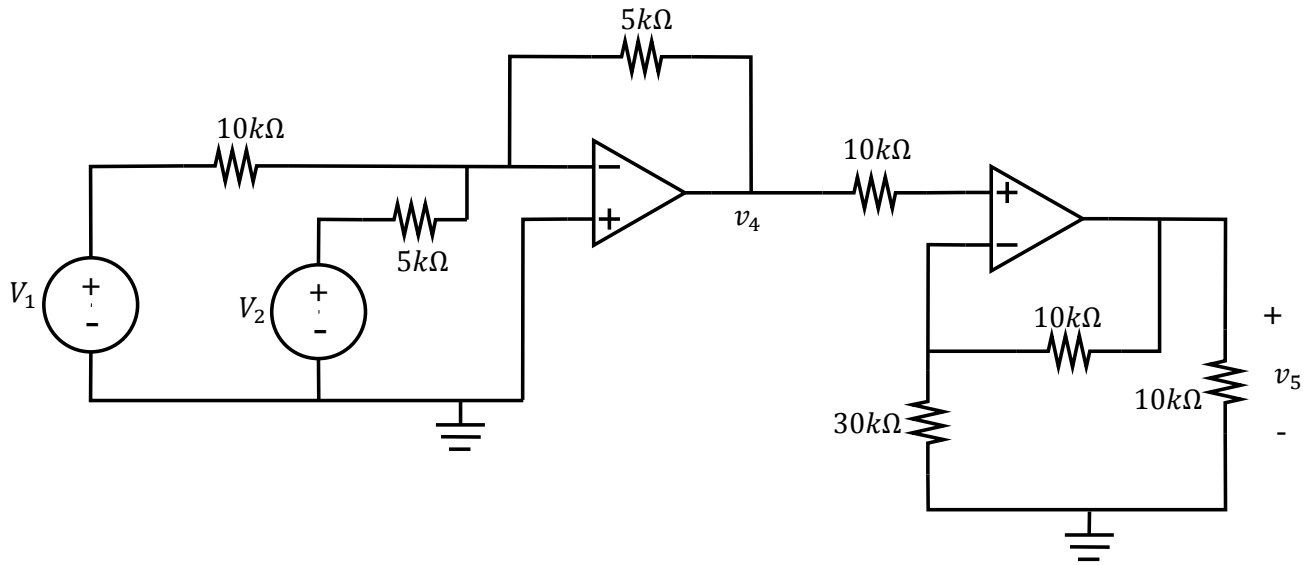


Vraag 2.4 [4]

- a) Bereken v_4 as $V_1 = 2V$ en $V_2 = 3V$.
b) Bepaal a in die volgende uitdrukking: $v_5 = av_4$.

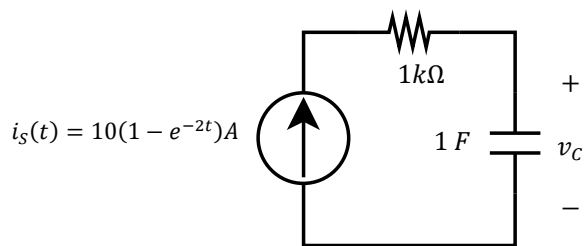
Question 2.4 [4]

- a) Determine v_4 if $V_1 = 2V$ and $V_2 = 3V$.
b) Find a in the following expression: $v_5 = av_4$.

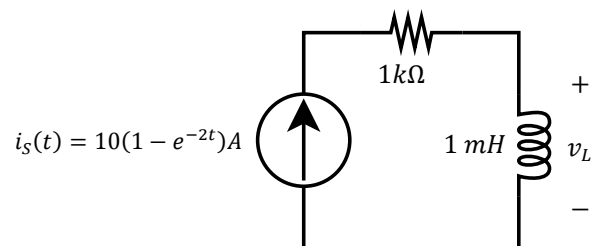


Vraag 2.5 [4]

- a) Bepaal v_C by $t = 1\text{s}$. Aanvaar $v_C(0) = -10\text{V}$.
b) Bepaal v_L by $t = 1\text{s}$.

**Question 2.5 [4]**

- a) Determine v_C at $t = 1\text{s}$. Assume $v_C(0) = -10\text{V}$.
b) Determine v_L at $t = 1\text{s}$.



Vraag 2.6 [2]

Aanvaar die baan is in stabiele-toestand en bereken die energie wat gestoor is in die $50\ \mu F$ kapasitor.

Question 2.6 [2]

Assume steady-state conditions and determine the energy stored in the $50\ \mu F$ capacitor.

